

WILEY



Review: Information Technology, E-Government, and Institutional Change

Reviewed Work(s): Building the Virtual State: Information Technology and Institutional Change by Jane E. Fountain

Review by: Stuart Bretschneider

Source: *Public Administration Review*, Nov. - Dec., 2003, Vol. 63, No. 6 (Nov. - Dec., 2003), pp. 738-741

Published by: Wiley on behalf of the American Society for Public Administration

Stable URL: <https://www.jstor.org/stable/3542469>

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at <https://about.jstor.org/terms>



JSTOR

Wiley and American Society for Public Administration are collaborating with JSTOR to digitize, preserve and extend access to *Public Administration Review*

Information Technology, E-Government, and Institutional Change

Stuart Bretschneider, Syracuse University

Jane E. Fountain, *Building the Virtual State: Information Technology and Institutional Change* (Washington, DC: Brookings Institution, 2001). 251 pp., \$44.95 cloth, ISBN: 0-8157-0078-4; \$18.95 paper, ISBN 0-8157-0077-6.

Over the past three decades, information technology has slowly but consistently permeated government organizations and institutions at all levels. This has largely been the result of an inevitable process of converting information from analog to digital forms. The lifeblood of government is information, and the digital revolution has allowed government organizations to more effectively and efficiently store, analyze, and retrieve information. This process has also been strongly affected by changes in telecommunications technology and the convergence of computer and communication technologies. The most recent manifestation of this process of technological change is the advent of the Internet or World Wide Web.

E-government, a catch-all phrase for use of the Internet by government to deliver services, collect data, and enhance democratic processes, has become the technological innovation du jour. Although there are several important new uses associated with e-government, such as direct delivery of services to citizens, it is, at its core, simply another information technology being adapted to governmental use. In that regard, many of the earlier, more general lessons of both tech-

nological and organizational change may apply to it. Yet, as has been true for each previous wave of new information technology, proponents of e-government claim it provides the mechanism to radically redesign governmental organizations and institutions (National Performance Review). Unfortunately, this sounds all too familiar. Almost 20 years ago a similar debate arose around public-management information systems, pitting proponents of new information technology against those who suggested that existing organizational and political relationships would dramatically influence any use of new technology (Bozeman and Bretschneider 1986; Kraemer and King 1986; Kraemer and Dedrick 1997).

Theory

Jane Fountain's book *Building the Virtual State: Information Technology and Institutional Change* clarifies the issues surrounding this debate and, more importantly, provides a powerful theoretical lens for understanding how technology, organizations, and institutions interact. The book is organized into two discrete sections. The first, entitled "Theory," covers a vari-

ety of topics within a somewhat unusual structure. Fountain begins by laying out her theoretical framework, entitled "technology enactment," which has three key elements. One is that objective forms of information technology are adjusted and manipulated once they are applied within some given organizational form. Second is that existing institutional arrangements both affect and are affected by those organizational forms. Finally, the resulting adopted and implemented form of information technology, called enacted technology, although a result of the first two processes, can send feedback that directly creates changes in the organization and indirectly in institutions through the organization. This framework operates at a somewhat high level of abstraction, but has the virtue of being applicable to any new technology, not just information or telecommunications technologies.

Once the theoretical framework is established, Fountain reviews recent reform efforts in the U.S. federal government, specifically the National Partnership for Reinventing Government, that rely to a large degree on the use of new information technologies. Reinvention argues that information

Stuart Bretschneider is the Laura J. and L. Douglas Meredith Professor for Teaching Excellence, as well as the director of the Center for Technology and Information Policy at Syracuse University's Maxwell School of Citizenship and Public Affairs. His research has focused on how public organizations make use of information technology and the effects of those technologies on public organizations; how public organizations employ forecasting technology and organize to carry out forecasting activities; and how sector differences affect administrative processes. Email: sibretsc@maxwell.syr.edu.

technology can be used to increase efficiency by changing organizations and institutions and developing the “virtual agency.” Reinvention is only the latest incarnation of the more generally recurring emphasis in public administration on governmental reform tied to managerial efficiency. Like its predecessors, reinvention assumes that organizations and institutions are essentially acted upon, being completely malleable and inert, a view that fails to account for the forces present within Fountain’s enacted technology model. Although Fountain accurately describes reinvention, it is clear that the underlying assumptions of how information technology will result in increased operational efficiency are overly simplistic. Reinvention comes to act as a sort of straw man argument throughout much of the rest of the book.

The next chapter reviews how computing affects organizations and shows how information technology is not a simple lever or knob for reengineering organizations and institutions. For example, the author notes how information technology can simultaneously bring about centralized control and decentralized action. Although the review does a good job of illustrating how information technology can have unanticipated impacts and even contradictory effects, it is not a review of the literature in this area. Rather it is focused more on core assumptions—like centralization, control, and the empowerment of workers—present in much of the reinvention literature.

The next two chapters review theories associated with bureaucracy and interorganizational networks, theories necessary for understanding the nature of organizations and institutions. One chapter reviews the basic tenets of the Weberian model of bureaucracy. Fountain suggests that certain aspects of this model may be outmoded and that alternative forms exist that could be more efficient and effective. This chapter also explores some of the implica-

tions of the Internet for Weberian bureaucracies. Some of the prior material on organizations informs this discussion, leading to propositions like “use of the Internet in bureaucracy is likely to lead to greater rationalization, standardization and use of rule-based systems.” Although some of these points seem contradictory, the essence of the argument is that new information technologies simultaneously support and supplant aspects of traditional bureaucracy. Again, these conclusions tend to flow from the complexity inherent in the enacted technology framework.

The following chapter on interorganizational networks has a similar feel to it. When reviewing the basic theory behind interorganizational networks, Fountain considers how information technologies like the Internet may affect the potential for such networks to become viable alternatives to traditional organizational bureaucracies. She also tries to integrate the concept of social capital into her analysis. For Fountain, social capital designates a noncorporeal resource (like goodwill) that is created from cooperative activities between people and organizations. Thus, interorganizational networks possess great potential for generating social capital. Consequently, if the Internet has some potential to enhance the viability and success of social networks, it has the capacity to enhance social capital. As in the previous chapter, though, applying the enacted technology framework does not automatically lead to a predictable result. All that can be said is that technological change does not automatically generate clearly anticipated organizational changes. Organizations and institutional arrangements will mediate the process, which often leads to unanticipated results.

The final chapter in the theory section emphasizes the role of institutions. Using the prior material on bureaucracies and technology, Fountain elaborates the enacted technology model

with a clear discussion about feedback and the recursive relationships between technology, organizations, and institutions. Clearly, Fountain’s theory is heavily influenced by Giddens’ work on structure and agency. Up to this point the theoretical framework has been quite general, but the final section on theory attempts to define how specific forms of web-based technology are expected to influence organizational and institutional arrangements. Four forms are suggested: agency Web sites, interagency Web sites, agency internal networks, and cross-agency integration and systems. These four forms are used first as illustrations of how the enacted technology theory can generate potentially testable hypotheses, and second as foci for the case studies presented in the second half of the book. The hypotheses presented at the end of the chapter are a mixed bag. Some, like proposition 3, “Agencies lack resources for learning to use IT,” although probably true, are not even hypotheses; others, like proposition 4, “Intergovernmental and public-private networks will over-shadow cross-agency IT-based networks because the institutional context favors those arrangements more readily than cross-agency federal networks,” are not clearly derived from either the framework or the four forms of virtual agency. Luckily the value of the theory and the book as a whole does not ride on these propositions.

Practice

The second half of the book presents three detailed cases: a cross-agency integration and systems (the International Trade Data System), an interagency Web site (the U.S. Business Advisor), and an agency internal network (the maneuver control system for the 9th Infantry Division). These cases are meant to illustrate how the enacted technology framework can be used to explain some of the complex interactions that occur when new in-

formation technologies are adopted. These chapters, some of the best and most interesting of the book, do an excellent job not only of illustrating the complex interrelationships suggested by the enacted technology framework but also of helping explain why technology adoption often results in unexpected outcomes.

In the case of the International Trade Data System (ITDS), by far the most complex of the three—requiring two chapters to present all the relevant details—Fountain first describes the complex array of federal, state, and local agencies involved in various aspects of international trade. One table identifies over 20 agencies (and more than 40 sublevel units) with some jurisdiction over international trade. Then she charts how international trade activities have changed since the 1970s, emphasizing the impact of these changes on the responsibilities of key agencies, especially the U.S. Customs Service. She shows that the volume of international trade impeded the Customs Service from carrying out its mission, and then focuses on the history of information automation within the agency. This history culminates with the requirement imposed by the North American Free Trade Agreement to harmonize trade transactions across North America. The need to fulfill this requirement, coupled with the advent of the reinvention movement, gave birth to the cross-agency concept of the ITDS. Her fascinating account of the effort through 2000 to implement the ITDS shows how, despite both the establishment of an interagency board to guide the implementation process and early claims about technological solutions, “One of the key information technology initiatives of the National Partnership for Reinventing Government appeared to have succumbed to the status quo” (144). Not only does this case clearly bring into question the idea of technological determinism, but it also solidly illustrates many of the core concepts

presented in the enacted technology framework. In fact the agency most threatened by the new system ends up gaining in importance.

The second case, though less ambitious, provides another excellent illustration of the enacted technology theory. The U.S. Business Advisor was designed to use web-based technology to link interagency services for small businesses through a single web portal, one that would provide one-stop shopping for small businesses dealing with the federal government. This case also shows how technological solutions derived from reinvention rhetoric do not result in straightforward organizational and institutional change. Unlike the ITDS, the U.S. Business Advisor Web site was launched and continues to function. The details of the case, though, demonstrate the difficulties of interagency cooperation and how grand objectives often end up being compromised. Besides the difficulties associated with interagency incentives and cooperation, politics played an important role: “For some technical specialists, the participation and lead of White House staff, however, dimmed their enthusiasm for what had been a technical, rather than political project” (155). Several other major portal-like Web site projects are briefly discussed at the end of the chapter, and Fountain suggests that some emerging Web sites and cross-linkages reflect interagency cooperation, but such activity remains limited. Given the extremely high rates of Web site construction and destruction, it seems fair to ask whether such interlinked Web sites in government will have greater staying power.

The final case focuses on the introduction of new information-based technology within a single organizational framework, the automation of the 9th Infantry Division of the U.S. Army during the 1980s. This case moves away from the current context of information systems and the World Wide Web to demonstrate the more general applica-

bility of the enacted technology model. The major finding in this case is that despite attempts simply to automate existing relationships, introducing new information technologies subtly changed roles and communications relationships. Nevertheless, Fountain notes this was not simply a case of technological determinism but rather a result of negotiated changes. Many of these negotiated changes resulted from specific existing institutions.

The book ends with a very short conclusion that focuses on the implications of the enacted technology model in three separate domains. First, Fountain considers the appropriate role for technology in modern institutional theory. She identifies a basic tension between, on the one hand, the worldview of reformism and reinvention, and on the other, the need for the stability of institutions. Even without an overarching summary, the book seems clearly to indicate that modern organizational change cannot be accomplished in a simplistic and mechanistic form. Next, the author considers the implications of her model for public management, the most salient implication being that networked organizations have their own logic, politics, and institutions. Although new forms of information technology are neither necessary nor sufficient for networked organizations, their existence enhances the likelihood that such networked structures will emerge. Consequently, Fountain recommends that public managers be prepared to function in such settings more often. This same point is made by the author in the third area, state building. Networked organizations as alternatives to older bureaucratic forms are likely to become more prevalent, which should suggest new strategies for the design of government organizations.

Strengths and Weaknesses

As each new wave of technological change arrives, advocates make

large, unsubstantiated claims about the power of the technology to reform and change all forms of human endeavor; this has been true in particular of information technology. Although some change usually occurs, the magnitude is generally modest and clearly channeled by the existing institutions. The recent history of the dot-com disaster suggests that this type of hype is not only incorrect, but may become dangerous if too many people invest too many resources in anticipation of the promised changes. *Building the Virtual State* provides a solid intellectual bulwark against these claims and, more importantly, against the potential bandwagon effect that could result. Thus, the major value of this book lies in the carefully articulated theory of enacted technology and its implications to public managers. But the theory alone would have little weight without the excellent case studies, which flesh out the general concepts and relationship. The cases will also help students of public administration understand why technology is not a panacea.

The major weakness of the book is that the enacted technology theory is highly abstract and generalized, making it difficult to use as a predictive tool. This is exemplified by the author's attempt to generate a few testable hypotheses from the theory, which ultimately fail. The cases help overcome this by providing concrete examples. A second weakness is that the book seems completely detached from prior work in public administration and information technology. As noted earlier, the work by Kraemer and associates dating back to the late 1970s and 1980s is extremely relevant: many of the findings from those earlier studies are completely compatible with Fountain's theory. A third weakness is the exclusive focus on cases from the U.S. federal government. Although information technology tends to be adopted first by larger organizations, by 2000 all states had formally adopted state-

level web portals and had designated a chief information officer. Even large county and city governments had made effective use of web technology by 2000. Finally, although Fountain acknowledges the continued relevance of Weberian bureaucracy, she tends to overstate the case for interorganizational alternatives. This is more a concern of emphasis. My personal view is that Weberian bureaucracy is still alive, well, and unlikely to yield its place to the much more fragile and unstable structures associated with interorganizational networks.

Building the Virtual State is already an influential book, and, I think, for all the right reasons. It provides the thoughtful administrator with a model for understanding technological change in general and information-technology-motivated change specifically. The cases put meat on the bones of the theory and also provide some excellent teaching material. I strongly recommend this book to anyone seriously interested in the role of technology as both cause and effect of organizational and institutional change.

References

- Bozeman, Barry, and Stuart Bretschneider. 1986. Public Management Information Systems: Theory and Prescription. *Public Administration Review* 46(Special Issue): 475–87.
- Kraemer, Kenneth, and John L. King. 1986. Computing and Public Organizations. *Public Administration Review* 46(Special Issue): 488–96.
- Kraemer, Kenneth, and Jason Dedrick. 1997. Computing in Public Organizations. *Journal of Public Administration Research and Theory* 7(1): 89–112.
- National Performance Review. 1993. *From Red Tape to Results: Creating a Government That Works Better and Cost Less*. Washington, DC: Government Printing Office.